

Implementing OTEL in FastAPI

A brief introduction to OTEL and how to instrument FastAPI applications.

Who am I ?

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Topics of interest

- Observability
- Kubernetes
- Microservices
- Distributed systems
- APIs design and governance
- Domain-Driven Design (DDD) and Event Driven Architecture's (EDA)
- DevOps and CI/CD
- Serverless computing
- Software Architecture

Topics to cover during the talk

- What's observability and which role plays OTEL?
- Three main pillars of observability
- Observability pillars and its particularities
- OTEL
 - Components
 - Instrumentation strategies
 - Collecting strategies
- How it looks like a good observability system?
- A simple FastAPI instrumentation to gather info and send it to Grafana using OTEL
- Q&A

What's observability?

Is the ability to understand a complex system's internal state by analyzing its external outputs (like logs, metrics and traces) to diagnose issues, optimize performance, and ensure reliability.

What are the three main pillars of observability?

Pillars of observability

- **Logs:** Structured records of events that provide context about system behavior.
- **Metrics:** Quantitative measurements that provide insights into system performance and health.
- **Traces:** Detailed records of requests as they flow through a system, showing the path and timing of each operation.

How it looks like a good observability system?

- **Semantic Convention (semconv):** Standardized naming and structure for logs, metrics, and traces to ensure consistency and interoperability.
- **Correlated Data:** The ability to link logs, metrics, and traces together for comprehensive analysis.
- **Contextual Information:** Enriching data with relevant context (e.g., user IDs, request IDs) to facilitate troubleshooting.
- **Real-time Monitoring:** The capability to monitor systems in real-time for immediate detection of issues.
- **Alerting:** The process of notifying relevant stakeholders when certain conditions are met, based on the analysis of logs, metrics, and traces.
- more...

What's OTEL?

OpenTelemetry (OTEL) is an open-source observability framework that provides a set of tools, APIs, and SDKs for collecting and exporting telemetry data (traces, metrics, and logs) from applications and services.

Why OTEL?

- Vendor-neutral standard for observability data.
- Comprehensive support for traces, metrics, and logs.
- Wide range of integrations with popular frameworks and libraries.
- Active community and ongoing development.
- Facilitates better monitoring, debugging, and performance optimization.
- Supports distributed tracing for microservices architectures.

Why semconv is important? - Semantic Conventions

- **Consistency:** Ensures that telemetry data follows a standardized format, making it easier to interpret and analyze.
- **Interoperability:** Facilitates integration between different observability tools and platforms.
- **Efficiency:** Reduces the effort required to instrument applications and services by providing predefined conventions.
- **Community Adoption:** Leverages widely accepted standards, promoting best practices in observability.
- **Cardinality Reduction:** Helps in reducing the number of unique metric names and labels, making it easier to manage and analyze metrics.
- more...

OTEL components

- **API:** Defines the interfaces for collecting telemetry data.
- **SDK:** Provides implementations of the API for different programming languages.
- **Exporters:** Components that send collected telemetry data to various backends (e.g., Jaeger, Prometheus, Zipkin).
- **Instrumentation libraries:** Pre-built libraries for popular frameworks and libraries to simplify the process of adding telemetry to applications.
- **Collector:** A separate service that can receive, process, and export telemetry data from multiple sources.

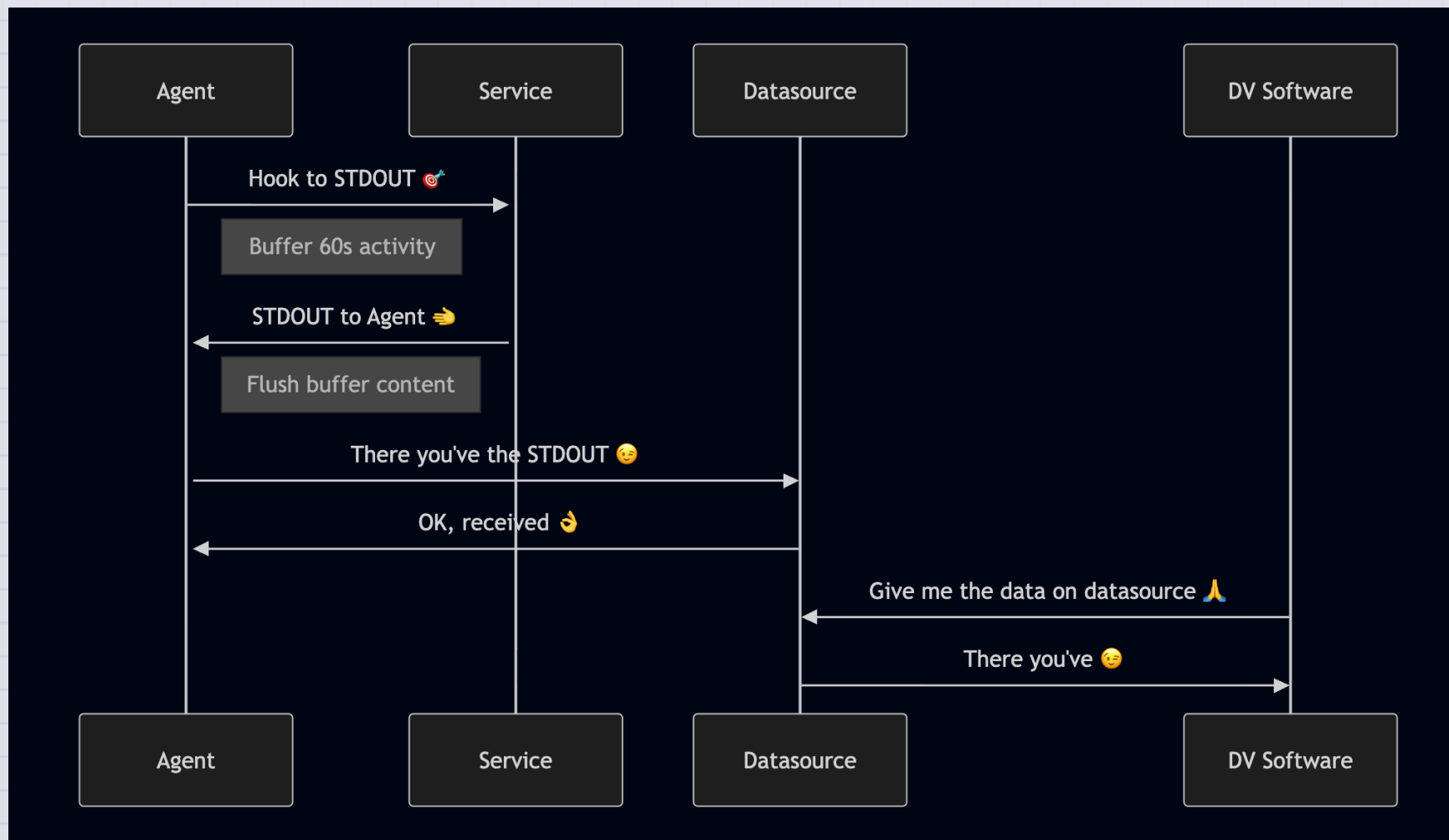
OTEL instrumentation strategies

- **Automatic Instrumentation:** Using pre-built instrumentation libraries to automatically collect telemetry data from popular frameworks and libraries without modifying application code.
- **Manual Instrumentation:** Adding custom instrumentation code to the application to collect specific telemetry data that is not covered by automatic instrumentation.

OTEL collecting strategies

- **Pull-based Collection:** The OTEL Collector periodically pulls telemetry data from instrumented applications or services using defined protocols.

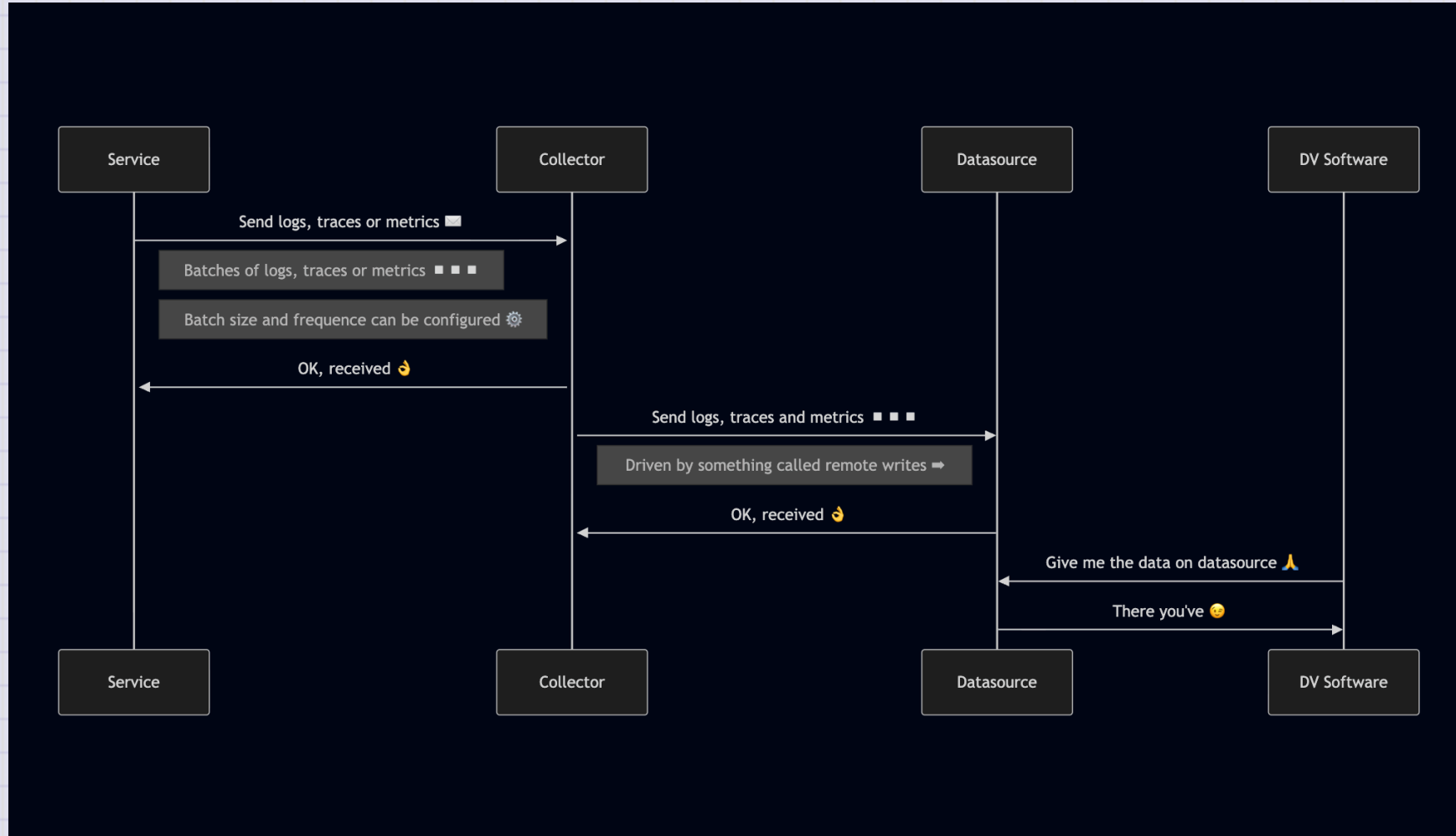
OTEL collecting strategies



OTEL collecting strategies

- **Push-based Collection:** Instrumented applications or services push telemetry data directly to the OTEL Collector or other backends.

OTEL collecting strategies



Typical OTEL collecting architecture

- Logs are collected through a pull-based strategy.
- Metrics and traces are collected through a push-based strategy.
- The OTEL Collector acts as a central hub for processing and exporting telemetry data.
- Data is sent to a backend observability platform such as Grafana
- Correlation between logs, metrics, and traces is achieved through consistent use of trace IDs and span IDs.

Which one is a good log format or structure?

01 - Plain text

```
[2023-10-01 12:00:00] INFO user-service v1.0.0 - user login successful | trace_id=4bf92f3577b34da6a3ce929d0e0e4736 span_id=00f067aa0ba902b7 user_id=12345 http_method=POST http_url=/login http_status_code=200 duration_ms=150
```

Which one is a good log format?

02 - JSON

```
{  
  "correlation_id": "4bf92f3577b34da6a3ce929d0e0e4736",  
  "timestamp": "2023-10-01T12:00:00Z",  
  "level": "INFO",  
  "service.name": "user-service",  
  "service.version": "1.0.0",  
  "trace_id": "4bf92f3577b34da6a3ce929d0e0e4736",  
  "span_id": "00f067aa0ba902b7",  
  "message": "user login successful",  
  "user.id": "12345",  
  "http.method": "POST",  
  "http.url": "/login",  
  "http.status_code": 200,  
  "duration_ms": 150  
}
```

Why JSON is better?

- **Structured Data:** JSON provides a structured format that is easy to parse and analyze.
- **Rich Context:** JSON allows for the inclusion of additional metadata and context, making it easier to correlate logs with traces and metrics.
- **Compatibility:** JSON is widely supported by logging systems and observability tools, facilitating integration and analysis.
- **Flexibility:** JSON can easily accommodate changes in log structure without breaking existing systems.
- **Searchability:** JSON logs can be easily indexed and searched in log management systems.
- more...

For what logs works?

- **Debugging:** Identifying and diagnosing issues in the application.
- **Auditing:** Tracking user actions and system changes for compliance and security.
- **Alerting:** Triggering notifications based on specific log events or patterns.
- **Usage Analytics:** Understanding user behavior and application usage patterns.
- more...

Traces

- **Tracing** is the process of tracking a full request through a distributed system, capturing detailed information about each operation (span) involved in processing a request.
- **Advantages:**
 - Provides a comprehensive view of request flow.
 - Helps identify performance bottlenecks.
 - Facilitates root cause analysis of issues.
- **Disadvantages:**
 - Can introduce overhead in terms of performance and storage.
 - Requires careful instrumentation to ensure meaningful data is collected.

Why traces over logs?

- **Holistic View:** Traces provide a complete picture of how requests flow through a system, while logs are often isolated events.
- **Performance Insights:** Traces can reveal latency and performance bottlenecks that may not be evident from logs alone.
- **Contextual Correlation:** Traces inherently link related operations, making it easier to understand the relationships between different components.
- **Reduced Noise:** Traces can help filter out irrelevant log data by focusing on specific requests or transactions.
- **End-to-End Visibility:** Traces offer visibility into the entire lifecycle of a request, from start to finish.
- more...

Metrics

- **Metrics** are quantitative measurements that provide insights into the performance and health of a system.
- **Why they are useful?**
 - Enable monitoring of system performance.
 - Facilitate capacity planning and resource management.
 - Help identify trends and anomalies in system behavior.
 - Support alerting based on predefined thresholds.
- **Methods to collect metrics?**
 - Local service metrics endpoint
 - Prometheus

Metrics meaning

- **Percentiles:** Statistical measures that indicate the value below which a given percentage of observations fall (e.g., 50th, 90th, 99th percentiles).
- **Averages:** The mean value of a set of measurements, providing a general overview of system performance.
- **Type of metrics:**
 - **Gauge** - represents a single numerical value that can go up and down in a specific point in time.
 - **Counter** - represents a cumulative value that only increases.
 - **Histogram** - samples observations and counts them in configurable buckets.
 - **UpDownCounter** - represents a value that can increase and decrease in a whole running application.

Minimal aspects of a metric

- **Name:** A unique identifier for the metric that describes what is being measured.
- **Kind:** The type of metric (e.g., Gauge, Counter, Histogram, UpDownCounter).
- **Unit:** The unit of measurement for the metric (e.g., seconds, bytes, requests).
- **Description (Optional):** A brief explanation of what the metric represents and its significance.

Instrumenting a FastAPI application with OTEL

Setting up the logger

```
import os
import logging

from opentelemetry.sdk.resources import Resource

from settings import Config
from observability.logger import init_otel_logger
from observability.config import OTLPConfig, LoggerConfig, LogLevel, otlp_headers_from_env

def configure_logger(resource: Resource, cfg: Config) -> logging.Logger:
    logger_config = LoggerConfig(
        resource=resource,
        otlp_config=OTLPConfig(
            endpoint=cfg.otel.otlp_endpoint,
            insecure=cfg.otel.otlp_insecure,
            headers=otlp_headers_from_env({}, cfg.otel.otlp_headers),
        ) if cfg.otel.log_exporter == "otlp" and cfg.otel.otlp_enabled else None,
        name=cfg.app.app_name,
        level=LogLevel.from_str(cfg.logger.level),
        formatter=logging.Formatter(os.getenv("OTEL_PYTHON_LOG_FORMAT", cfg.logger.format)),
    )

    return init_otel_logger(logger_config)
```

Instrumenting a FastAPI application with OTEL

Setting up the traces provider

```
import os
import logging

from opentelemetry.sdk.resources import Resource
from opentelemetry.sdk.trace import TracerProvider

from settings import Config
from observability.trace_provider import init_otel_trace_provider
from observability.config import OTLPConfig, TracerConfig, otlp_headers_from_env


def configure_trace_provider(resource: Resource, cfg: Config) -> TracerProvider:
    tracer_config = TracerConfig(
        resource=resource,
        otlp_config=OTLPConfig(
            endpoint=cfg.otel.otlp_endpoint,
            insecure=cfg.otel.otlp_insecure,
            headers=otlp_headers_from_env({}, cfg.otel.otlp_headers),
            batched=cfg.otel.traces_batched,
        ) if cfg.otel.otlp_enabled and cfg.otel.exporter == "otlp" else None,
    )

    return init_otel_trace_provider(tracer_config)
```

Instrumenting a FastAPI application with OTEL

Setting up the metrics provider


```

import os
import logging

from fastapi import FastAPI
from opentelemetry.sdk.resources import Resource
from opentelemetry.sdk.metrics import MeterProvider

from settings import Config
from metrics_router import metrics_router
from observability.meter_provider import init_otel_meter_provider
from observability.config import OTLPConfig, MeterConfig, otlp_headers_from_env

def configure_meter_provider(app_instance: FastAPI, resource: Resource, cfg: Config) -> MeterProvider:
    meter_config = MeterConfig(
        resource=resource,
        otlp_config=OTLPConfig(
            endpoint=cfg.otel.otlp_endpoint,
            insecure=cfg.otel.otlp_insecure,
            headers=otlp_headers_from_env({}, cfg.otel.otlp_headers),
        ) if cfg.otel.metric_exporter == "otlp" and cfg.otel.otlp_enabled else None,
        enable_prometheus_reader=cfg.otel.prometheus_enabled,
    )

    meter_provider, prometheus_reader = init_otel_meter_provider(meter_config)
    if prometheus_reader is not None:
        app_instance.include_router(metrics_router)

    return meter_provider

```

Instrumenting a FastAPI application with OTEL

Setting up the FastAPI app with OTEL

```
def create_app() -> FastAPI:
    cfg = get_config()
    app_instance = FastAPI(title=cfg.app.doc_title, version=cfg.app.app_version)

    resource = ResourceBuilder.new(). \
        with_name(cfg.app.app_name). \
        with_version(cfg.app.app_version). \
        with_namespace(cfg.otel.service_namespace). \
        with_environment(cfg.app.environment). \
        build()

    trace_provider = configure_trace_provider(resource, cfg)
    meter_provider = configure_meter_provider(app_instance, resource, cfg)
    logger = configure_logger(resource, cfg)

    FastAPIInstrumentor().instrument_app(
        app=app_instance,
        tracer_provider=trace_provider,
        meter_provider=meter_provider,
    )
    logger.info("service started...")
    return app_instance
```

Adding a prometheus endpoint to expose metrics

```
def configure_meter_provider(app_instance: FastAPI, resource: Resource, cfg: Config) -> MeterProvider:
    ... # previous code
    meter_provider, prometheus_reader = init_otel_meter_provider(meter_config)
    if prometheus_reader is not None:
        app_instance.include_router(metrics_router)
    ... # rest of the code
```

```
from fastapi import APIRouter, Response
from prometheus_client import generate_latest, CONTENT_TYPE_LATEST
```

```
metrics_router = APIRouter()
```

```
@metrics_router.get("/metrics")
def get_metrics(response: Response):
    response.headers["Content-Type"] = CONTENT_TYPE_LATEST
    return generate_latest()
```

Which kind of things we can monitor with OTEL in FastAPI?

- **HTTP Requests:** Monitor incoming requests, response times, status codes, and error rates.
- **Database Queries:** Track database interactions, query performance, and error rates.
- **External API Calls:** Monitor calls to third-party services, including latency and failure rates.
- **Custom Business Logic:** Instrument specific business processes or workflows to gather performance metrics and traces.
- **Resource Utilization:** Collect metrics on CPU, memory, and other resource usage of the application.
- more...

Q&A

- Thank you for your attention!
- Any questions?

Get the source code

